

Robust Identity Testing for Balanced High-Temperature Ising Models with Subcube Conditioning

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We study identity testing of an arbitrary distribution P on $\{-1, +1\}^n$ against an explicitly specified Ising distribution Q , using subcube-conditional samples from P . The partition function and target marginals are not given. We prove that the optimal $\tilde{O}(\sqrt{n}/\varepsilon^2)$ dependence on dimension and distance extends from the uniform target to every Dobrushin-unique Ising target whose one-site conditionals remain in $[\eta, 1 - \eta]$ after arbitrary pinnings and projections. More precisely, with Dobrushin gap κ , our adaptive tester uses

$$O\left(\kappa^{-6}\eta^{-8}\frac{\sqrt{n}}{\varepsilon^2}\log^{40}\frac{16n}{\varepsilon\kappa\eta\delta}\right)$$

hidden-oracle calls and errs with probability at most δ . The hidden distribution is otherwise unrestricted. Every nontrivial restricted query is supported by a previously sampled hidden anchor (and the all-star query is intrinsically supported), so the algorithm never uses an undefined zero-mass conditional. The proof develops a field-uniform heat-bath robust-Pisier inequality, a target-relative random-restriction theorem, and a target-paired tensor mean test. Pinned target samples are generated approximately by Glauber dynamics with transcript-level error accounting. A matching $\Omega(\sqrt{n}/\varepsilon^2)$ adaptive-SCOND lower bound holds for balanced product Ising alternatives. The balance parameter is an explicit quantitative parameter; no balance-free, field-uniform claim is made under Dobrushin uniqueness alone.

Additional Key Words and Phrases: distribution testing, Ising model, subcube conditioning, Dobrushin uniqueness, random restrictions, robust Pisier inequality

1 INTRODUCTION

Conditional sampling can expose local structure that ordinary samples hide. For high-dimensional binary distributions, the subcube oracle (SCOND) is especially natural: a query fixes an arbitrary subset of coordinates and receives a sample from the resulting face. Canonne, Chen, Kamath, Levi, and Waingarten (CCKLW) proved that uniformity can be tested with $\tilde{O}(\sqrt{n}/\varepsilon^2)$ SCOND queries [6, Theorems 1.1 and 2.1]. The mechanism is a random-restriction dichotomy: global TV distance is found either in a projection or in a smaller supported face, and projection distance is converted into a detectable conditional-mean signal.

This paper asks how far that mechanism extends when the known target is a dependent, nonuniform Gibbs law. The visible target is

$$Q(x) = q_{J,h}(x) = \frac{1}{Z_{J,h}} \exp\left(\sum_{i<j} J_{ij}x_i x_j + \sum_i h_i x_i\right), \quad x \in \{-1, +1\}^n, \quad (1)$$

and the unknown P is arbitrary. We assume Dobrushin uniqueness. With $C_{ij} = \tanh |J_{ij}|$ for $i \neq j$ and $C_{ii} = 0$, the gap is

$$\|C\|_1 = \|C\|_\infty \leq 1 - \kappa. \quad (2)$$

Symmetry of J makes C symmetric, so the two displayed norms agree.

The main obstacle is not weak dependence but rare target mass. Condition (2) controls interactions and says nothing about the fields h_i . A target coordinate can therefore be arbitrarily close to deterministic. Uniform-target arguments that identify a likelihood drop with an unweighted mean residual then lose an arbitrarily large factor. We isolate a quantitative sufficient condition under which the square-root query bound can be proved: a hereditary lower and upper bound on target one-site conditionals. The dependence on this balance parameter is displayed throughout.

Main contribution. For every promised valid rational lower bound $\eta \in (0, 1/2]$, we give a robust identity tester for all Dobrushin Ising targets that are hereditarily η -balanced. It uses

$$O\left(\kappa^{-6}\eta^{-8}\frac{\sqrt{n}}{\varepsilon^2}\log^{40}\frac{16n}{\varepsilon\kappa\eta\delta}\right)$$

SCOND calls to P . No structural promise is imposed on P . The algorithm does not evaluate $Z_{J,h}$, any target marginal, or an exact target sampler. It uses only explicit one-site logistic probabilities and an approximate pinned-target Glauber sampler. Approximation is coupled at the level of the full adaptive transcript.

The dependence on n and ε is tight up to logarithms; no optimality is claimed for the displayed κ, η exponents. We prove a self-contained $\Omega(\sqrt{n}/\varepsilon^2)$ lower bound against adaptive SCOND algorithms using full-support product Ising alternatives with balance at least $1/4$.

Proof architecture. The proof has three new components.

- (i) A directed heat-bath robust-Pisier inequality with constant $O(\kappa^{-3/2}\log(n+1))$, uniform over external fields, pinnings, and coordinate projections.
- (ii) A relative random-restriction theorem. If a projection has TV distance α , then hidden-anchored restrictions of dimensions t and $t+1$ have expected conditional-mean residual at least $(t/n)\alpha$ divided by an explicit $\kappa^{-5/2}\eta^{-4}\text{polylog}(n/\alpha)$ factor.
- (iii) A target-paired tensor test. Pairing $X \sim P$ with $Y \sim Q$ makes the mean $\mathbb{E}(X - Y)$ observable without knowing $\mathbb{E}Q$. A tensor blow-up argument handles the arbitrary covariance of P ; Dobrushin concentration is needed only under the null.

These components are combined by a recursive work-investment scheme in the style of CCKLW, but all target-relative steps and computational primitives are proved here.

Scope. The balance assumption belongs to the target only. It is inherited by every face and projection used by the algorithm. The paper makes no balance-free, field-uniform claim under Dobrushin uniqueness alone. Every finite field vector does induce a positive balance parameter, but it may be exponentially small in the input fields. In particular, this is not the easier problem of uniformity testing under an Ising promise on the hidden law: our hidden law is unrestricted and our target is generally nonuniform and dependent.

Organization. Section 2 fixes the oracle and zero-mass conventions. Section 3 states the main and structural theorems. Sections 4–6 give the tester, correctness, query recurrence, and implementation. The complete new proofs appear in Appendices A–C; Appendix D proves the finite standard lemmas in the normalizations used here, and Appendix E proves the lower bound.

2 MODEL AND PRELIMINARIES

We write spins in $\{-1, +1\}$ and use bits $B_i = (X_i + 1)/2$ when discussing mean vectors. For distributions P, Q on a finite set,

$$d_{\text{TV}}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|.$$

A restriction is $\rho \in \{-1, +1, *\}^n$. Its free set is $V(\rho) = \{i : \rho_i = *\}$ and $m(\rho) = |V(\rho)|$. If $P[\rho] > 0$, P^ρ denotes P conditioned on all pins in ρ . We never write or query P^ρ before positivity has been established.

2.1 The strict SCOND convention

An SCOND query is a restriction ρ with $P[\rho] > 0$ and returns an independent sample from P^ρ . If $P[\rho] = 0$, the conditional law is undefined; the oracle supplies no flag, sample, or other information. Every query made below is legal for the following reason.

Definition 2.1 (Anchored restriction law). For a supported current face ρ and $0 \leq t \leq m(\rho)$, the law $\mathcal{D}_{P^\rho}(t)$ is generated by drawing an anchor $Y \sim P^\rho$, choosing a uniform t -subset $T \subseteq V(\rho)$, and pinning every coordinate in $V(\rho) \setminus T$ to its value in Y .

The resulting child contains Y , hence has positive P^ρ -mass. This argument is conditional on the complete past transcript and therefore remains valid under adaptive recursion. Analytic proofs may compare likelihoods at unqueried endpoints, but the algorithm never queries such an endpoint.

2.2 Pinning and balance

Pinning an Ising model deletes the pinned rows and columns of J and changes the field on a free coordinate to

$$h_i^\rho = h_i + \sum_{j: \rho_j \neq * } J_{ij} \rho_j. \quad (3)$$

Consequently the active influence matrix is a principal submatrix of C and retains Dobrushin gap at least κ .

Definition 2.2 (Hereditary balance). A strictly positive target Q is hereditarily η -balanced if for every pin ρ , every $A \subseteq V(\rho)$, every $i \in A$, and every feasible $z \in \{-1, +1\}^{A \setminus \{i\}}$,

$$\eta \leq Q(B_i = 1 \mid \rho, B_{A \setminus \{i\}} = z) \leq 1 - \eta.$$

The condition explicitly includes one-site conditionals in every coordinate projection. It has a simple visible certificate.

PROPOSITION 2.3 (BOUNDED LOCAL FIELDS IMPLY BALANCE). *Put*

$$H_* := \max_i \left(|h_i| + \sum_{j \neq i} |J_{ij}| \right).$$

Then Q is hereditarily η -balanced for $\eta = (1 + e^{2H_})^{-1}$. If instead $\max_i |h_i| \leq H$ and (2) holds, one may certify*

$$\eta \geq \left(1 + e^{2H + 2 \arctanh(1-\kappa)} \right)^{-1}.$$

PROOF. With the full boundary exposed,

$$Q(X_i = +1 \mid x_{-i}) = \left(1 + e^{-2(h_i + \sum_{j \neq i} J_{ij} x_j)} \right)^{-1},$$

which lies in $[\eta, 1 - \eta]$. A conditional with a smaller sigma-field is an average of these full-boundary conditionals, so the same interval is preserved. This also proves inheritance under pins and projections. For the second assertion, if $c_j = \tanh |J_{ij}|$ then $\sum_j c_j \leq 1 - \kappa$ and monotonicity of $\arctanh(x)/x$ gives

$$\sum_j |J_{ij}| = \sum_j \arctanh(c_j) \leq \arctanh\left(\sum_j c_j\right) \leq \arctanh(1 - \kappa).$$

□

The theorem takes promised valid rational lower bounds κ and η as input. The tester need not verify the hereditary definition itself. For rational or interval-valued (J, h) , conservative certificates can be obtained by outward-rounded interval arithmetic. This numerical verification is separate from the statistical promise.

2.3 Residuals and projection distance

For a supported face ρ , define the bit-mean residual

$$\Delta(\rho)_i = P^\rho(B_i = 1) - Q^\rho(B_i = 1), \quad i \in V(\rho). \quad (4)$$

For a current m -coordinate instance and $0 \leq t \leq m$, define

$$\alpha_t = \mathbb{E}_{T:|T|=t} d_{\text{TV}}(P_{T^c}, Q_{T^c}), \quad \beta_t = \mathbb{E}_{\rho' \sim \mathcal{D}_P(t)} d_{\text{TV}}(P^{\rho'}, Q^{\rho'}). \quad (5)$$

All notation in (5) is relative to the current supported face; we suppress the superscript ρ .

The following exact TV direction drives the recursion.

LEMMA 2.4 (PROJECTION-RESTRICTION INEQUALITY). *For fixed T ,*

$$\begin{aligned} d_{\text{TV}}(P, Q) &\leq d_{\text{TV}}(P_{T^c}, Q_{T^c}) \\ &\quad + \mathbb{E}_{Z \sim P_{T^c}} d_{\text{TV}}(P(X_T \in \cdot \mid Z), Q(X_T \in \cdot \mid Z)). \end{aligned}$$

Consequently $d_{\text{TV}}(P, Q) \leq \alpha_t + \beta_t$.

PROOF. In the ℓ_1 sum for $P(z, x) - Q(z, x)$, add and subtract $P_{T^c}(z)Q(x \mid z)$ and use the triangle inequality. Summing the first term over x gives the projection TV; the second is averaged under P_{T^c} . Average over T . No KL divergence or Pinsker inequality is used. $\square$

3 RESULTS AND STRUCTURAL INGREDIENTS

3.1 Main theorem

THEOREM 3.1 (BALANCED ROBUST IDENTITY TESTING). *Let the visible Ising target $Q = q_{J,h}$ satisfy $\|C\|_1 = \|C\|_\infty \leq 1 - \kappa$, and let a lower bound $\kappa \in (0, 1]$ and a promised valid rational lower bound $\eta \in (0, 1/2]$ be supplied such that Q is hereditarily η -balanced. For every arbitrary hidden law P , every $\varepsilon \in (0, 1)$, and every $\delta \in (0, 1/3)$, there is an adaptive binary-SCOND tester distinguishing*

$$P = Q \quad \text{from} \quad d_{\text{TV}}(P, Q) \geq \varepsilon$$

with error at most δ and at most

$$M = C\kappa^{-6}\eta^{-8}\frac{\sqrt{n}}{\varepsilon^2} \left[\log \frac{16n}{\varepsilon\kappa\eta\delta} \right]^{40} \quad (6)$$

hidden-oracle calls, for a universal C .

The tester uses neither $Z_{J,h}$ nor exact target marginals. Exact target sampling is unnecessary: an implementation using approximate pinned Glauber samples has the same hidden-query bound and total error at most δ . Every nontrivial restricted hidden conditional query is supported by a preceding hidden anchor; the all-star query is intrinsically supported.

COROLLARY 3.2 (KNOWN PRODUCT TARGETS). *Let $J = 0$ and write $q_i = Q(X_i = +1)$. If*

$$\eta \leq \min_{i \in [n]} \min\{q_i, 1 - q_i\},$$

then Theorem 3.1 applies with $\kappa = 1$ against an arbitrary hidden law, using

$$O\left(\eta^{-8}\frac{\sqrt{n}}{\varepsilon^2} \log^{40} \frac{16n}{\varepsilon\eta\delta}\right)$$

binary-SCOND queries. Thus arbitrary known biased products are genuinely covered; $q_i = 1/2$ recovers uniformity testing up to polylogarithmic factors.

The proof rests on the next three results. Their complete proofs appear in the appendices.

3.2 Field-uniform heat-bath robust Pisier

For a law μ on a binary cube, let X^i denote the state with coordinate i flipped. Conditional on X , let H_i be independent Bernoulli variables whose parameters are the heat-bath flip probabilities $\mu(X_i = -x_i \mid X_{-i} = x_{-i})$ at the realized state $X = x$. An orientation selects one direction of every undirected cube edge.

THEOREM 3.3 (PINNED AND PROJECTED ROBUST PISIER). *Every pinned Dobrushin Ising target on m active coordinates, and every coordinate projection of such a target, satisfies*

$$\mathbb{E}_\mu |F - \mathbb{E}_\mu F| \leq K_m \mathbb{E}_{X, H, \epsilon} \left| \sum_i \epsilon_i H_i \mathbf{1}\{X \rightarrow X^i\} (F(X) - F(X^i)) \right|, \quad (7)$$

for every real F and every orientation, where

$$K_m \leq \frac{C}{\kappa} \sqrt{1 + \frac{1 - \kappa}{\min\{\tanh(1/4), \kappa/2\}}} \log(m+1) \leq C\kappa^{-3/2} \log(m+1). \quad (8)$$

The constant is independent of all external fields.

The proof combines a Gaussian product-mixture representation in a small-interaction regime, a weighted defective-block reduction, and a new projection-transfer argument based on concavity of stationary heat-bath flow.

3.3 Balanced relative random restrictions

For a target projection define the conditional influence radius

$$V_m = \sup_{\rho, i} \left\| \left(Q(B_j = 1 \mid \rho, B_i = 1) - Q(B_j = 1 \mid \rho, B_i = 0) \right)_{j \neq i} \right\|_2. \quad (9)$$

Dobrushin comparison gives $V_m \leq (1 - \kappa)/\kappa$.

Set

$$\begin{aligned} c_\eta &= \eta(1 - \eta), \quad L_\eta = 3 \frac{1 - \eta}{\eta}, \quad B_\eta = 3 \left(\frac{1 - \eta}{\eta} \right)^2, \quad D_V = 8(1 + V_m), \\ H_a &= 2 + \left\lceil \log_2 \frac{4}{a} \right\rceil, \quad R_a = 2 + \left\lceil \log_2 \frac{64 K_m L_\eta \sqrt{m}}{c_\eta a} \right\rceil. \end{aligned} \quad (10)$$

THEOREM 3.4 (BALANCED RELATIVE RESTRICTION). *For a current supported m -coordinate instance, if $1 \leq t < m$ and $\alpha_t \geq a$, then*

$$\begin{aligned} &\mathbb{E}_{\mathcal{D}_P(t)} \|\Delta\|_2 + \mathbb{E}_{\mathcal{D}_P(t+1)} \|\Delta\|_2 \\ &\geq c \frac{t}{m} \frac{c_\eta \alpha_t}{K_m L_\eta B_\eta D_V H_a R_a^2 \log(2m)}. \end{aligned} \quad (11)$$

In particular,

$$\mathbb{E}_{\mathcal{D}_P(t)} \|\Delta\|_2 + \mathbb{E}_{\mathcal{D}_P(t+1)} \|\Delta\|_2 \geq \frac{t}{m} \frac{\alpha_t}{\Lambda(m, a)}, \quad (12)$$

where

$$\Lambda(m, a) \leq C\kappa^{-5/2} \eta^{-4} \log^5 \frac{16m}{a\kappa\eta}. \quad (13)$$

The two adjacent dimensions are essential to the incidence argument. The factor t/m is obtained by exact double counting, not by changing from a target-anchored to a hidden-anchored restriction law.

3.4 A target-paired arbitrary-alternative mean test

THEOREM 3.5 (TARGET-PAIRED RAW-MEAN TEST). *Let P be arbitrary on a supported m -coordinate face and let Q be a pinned Dobrushin target. For $0 < u \leq \sqrt{m}$ there is a test $\text{RAWMEAN}(\rho, u, \xi)$ of*

$$P^\rho = Q^\rho \quad \text{versus} \quad \|\Delta(\rho)\|_2 \geq u$$

with error at most ξ , using at most

$$C \max \left\{ 1, \frac{\sqrt{m}}{\kappa} \max\{u^{-2}, u^{-1}\} \right\} \log^3 \frac{16m}{u\xi} \quad (14)$$

hidden samples and the same number of internally generated visible samples. For $u > \sqrt{m}$ the test accepts immediately, since the alternative is empty.

For paired spins $X \sim P^\rho$, $Y \sim Q^\rho$, set $W = (X - Y)/2$. Then $\mathbb{E}W = \Delta(\rho)$ in bit notation. The test uses split empirical inner products of $W^{\otimes 2^k}$; it never materializes a tensor. Large alternative variance forces signal growth at the next tensor level, so no covariance promise on P is present.

3.5 Optimality in dimension and distance

THEOREM 3.6 (ADAPTIVE-SCOND LOWER BOUND). *There are universal $c, C, \varepsilon_0 > 0$ such that for $0 < \varepsilon \leq \varepsilon_0$ and $n \geq C/\varepsilon^2$, every adaptive randomized SCOND identity tester with error at most $1/3$ needs $c\sqrt{n}/\varepsilon^2$ queries, even when the visible target is uniform and every alternative is a full-support product Ising law with $J = 0$ and hereditary balance at least $1/4$.*

Thus the n, ε dependence in Theorem 3.1 is optimal up to logarithms. We make no optimality claim for its explicit $\kappa^{-6}\eta^{-8}$ dependence.

4 THE TESTER

This section gives the literal recursive tester. The constants $C_L, C_N, C_M, C_\sigma, C_R, C_{\text{alloc}}$ are fixed positive integers, chosen in this order to satisfy the inequalities in Section 5.2. They depend on no input parameter. All logarithms in the algorithm are base two; changing the base changes only these integers.

For $0 < a \leq b$, let

$$\text{Dyad}(a, b) = \{\min(2^j a, b) : 0 \leq j \leq \lceil \log_2(b/a) \rceil\}, \quad (15)$$

with duplicates removed. At the root define

$$\ell_0 = \max \left\{ 2, \left\lceil \log_2 \frac{2^{20} n^3}{\varepsilon \kappa \eta \delta} \right\rceil \right\}, \quad \mathcal{L} = C_L \ell_0, \quad (16)$$

$$B_{\text{alloc}} = \lceil C_{\text{alloc}} \kappa^{-6} \eta^{-8} \sqrt{n} \varepsilon^{-2} \ell_0^{80} \rceil, \quad \delta_* = \frac{\delta}{100 B_{\text{alloc}}}, \quad \sigma = \frac{1}{C_\sigma \mathcal{L}^8}. \quad (17)$$

The allocation is fixed before the recursion; it is not defined in terms of the realized call tree.

PROPOSITION 4.1 (ORACLE LEGALITY). *Every hidden-oracle call in Algorithms 1–3 is a legal SCOND query, even conditional on the full adaptive history. No information is obtained from a zero-mass restriction.*

Algorithm 1 SUPPORTEDFACE(ρ, s)

-
- 1: Query the hidden oracle on the supported current face ρ and obtain an anchor x .
 - 2: Choose a uniform s -subset $S \subseteq V(\rho)$.
 - 3: Let ρ' extend ρ by setting $\rho'_i = x_i$ for every $i \in V(\rho) \setminus S$.
 - 4: **return** ρ' .
-

Algorithm 2 COREID(ρ, e)

-
- 1: $m \leftarrow |V(\rho)|$.
 - 2: **if** $m = 1$ **then**
 - 3: **return** the decision of RAWMEAN(ρ, e, δ_*).
 - 4: $t \leftarrow \max\{1, \lfloor \sigma m \rfloor\}$.
 - 5: $\bar{\Lambda} \leftarrow C_N \kappa^{-5/2} \eta^{-4} \mathcal{L}^5$ and $g \leftarrow (t/m)e / (8\bar{\Lambda})$.
 - 6: **for** $s \in \{t, t+1\}$ with $s \leq m$ **do**
 - 7: **for** $u \in \text{Dyad}(g/2, \sqrt{s})$ **do**
 - 8: $N_u \leftarrow \lceil C_N(u/g) \mathcal{L} \rceil$.
 - 9: **for** $r = 1, \dots, N_u$ **do**
 - 10: $\rho' \leftarrow \text{SUPPORTEDFACE}(\rho, s)$.
 - 11: **if** RAWMEAN(ρ', u, δ_*) rejects **then**
 - 12: **return** REJECT.
 - 13: **for** $v \in \text{Dyad}(e/4, 1)$ **do**
 - 14: $M_v \leftarrow \lceil C_M(v/e) \mathcal{L} \rceil$.
 - 15: **for** $r = 1, \dots, M_v$ **do**
 - 16: $\rho' \leftarrow \text{SUPPORTEDFACE}(\rho, t)$.
 - 17: **if** COREID(ρ', v) rejects **then**
 - 18: **return** REJECT.
 - 19: **return** ACCEPT.
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Algorithm 3 BALANCEDISINGID($J, h, \kappa, \eta, \varepsilon, \delta$)

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- 1: $R \leftarrow \lceil C_R \log_2(4/\delta) \rceil$.
 - 2: Independently run COREID($*^n, \varepsilon$) exactly R times.
 - 3: **return** REJECT iff at least $R/2$ runs reject.
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PROOF. The all-star face is supported. Inductively, assume the current ρ is supported. The first call in Algorithm 1 is therefore legal. Its child ρ' contains the sampled x , so $P^\rho[\rho'] > 0$. All calls made by RAWMEAN or a recursive invocation on ρ' are legal. The induction is conditional on the past, so it also covers arbitrary adaptive branch choices. The algorithm never submits an independently proposed face and never observes a zero-mass failure symbol. $\square$

The one-dimensional base case is legitimate because on one bit $d_{\text{TV}}(P^\rho, Q^\rho) = |\Delta(\rho)|$. Using RAWMEAN avoids computing an exact target Bernoulli marginal.

5 CORRECTNESS AND QUERY COMPLEXITY

We first record the elementary work-investment statement used twice.

LEMMA 5.1 (DYADIC WORK INVESTMENT). *Let $0 \leq U \leq M$ and $\mathbb{E}U \geq 2g > 0$. Then some $u \in \text{Dyad}(g, M)$ satisfies*

$$\mathbb{P}(U \geq u) \geq \frac{cg}{u[1 + \log_2(M/g)]} \quad (18)$$

for a universal $c > 0$.

PROOF. Let $u_j = \min(2^j g, M)$ and $J = \lceil \log_2(M/g) \rceil$. If the conclusion failed at every u_j , the layer-cake decomposition would give

$$\mathbb{E}U \leq g + \sum_{j=0}^{J-1} (u_{j+1} - u_j) \mathbb{P}(U \geq u_j) < g + \sum_{j=0}^{J-1} u_j \frac{cg}{u_j(J+1)} < 2g$$

for any fixed $c \leq 1/2$, a contradiction. $\square$

5.1 Completeness and soundness

LEMMA 5.2 (ONE-CORE CORRECTNESS). *The universal constants in Algorithms 2–3 can be fixed so that an ideal-target invocation $\text{COREID}(\rho, e)$ has the following properties.*

- (1) *If $P^P = Q^P$, it accepts except with probability at most $B(\rho, e)\delta_*$, where $B(\rho, e)$ is the deterministic number of fallible primitive calls below that node.*
- (2) *If $d_{\text{TV}}(P^P, Q^P) \geq e$, it rejects with probability at least $2/3$.*

PROOF. The first assertion follows by induction: equality is preserved under every supported pin, and each RAWMEAN invocation has error at most δ_* . A union bound charges one error unit to each fallible primitive.

For soundness use strong induction on $m = |V(\rho)|$. The case $m = 1$ is Theorem 3.5. Let $m > 1$, choose t as in the algorithm, and use Lemma 2.4. Either $\alpha_t \geq e/2$ or $\beta_t \geq e/2$.

In the first case, Theorem 3.4, with $a = e/2$ and $\bar{\Lambda}$ larger than its bound, gives

$$\mathbb{E}_{\mathcal{D}_P(t)} \|\Delta\|_2 + \mathbb{E}_{\mathcal{D}_P(t+1)} \|\Delta\|_2 \geq 4g.$$

The equal mixture of the two levels therefore has mean at least $2g$ and is bounded by $\sqrt{t+1}$. Lemma 5.1 supplies a tested level u whose occurrence probability is $\Omega(g/(u\mathcal{L}))$. Since $N_u = \Theta((u/g)\mathcal{L})$, a corresponding face is sampled with constant probability; conditional on that event, RAWMEAN rejects with probability at least $1 - \delta_*$. Choosing C_N sufficiently large makes this branch reject with probability at least $5/6$.

In the second case apply Lemma 5.1 to $U = d_{\text{TV}}(P^{\rho'}, Q^{\rho'}) \in [0, 1]$. There is a $v \in \text{Dyad}(e/4, 1)$ with

$$\mathbb{P}(U \geq v) \geq \frac{ce}{v \log_2(4/e)}.$$

Every recursive child has strictly smaller dimension: if $\sigma m \geq 2$ then $t \leq \sigma m \leq m/2$, while otherwise $t = 1 \leq m/2$. By the induction hypothesis a v -far child rejects with probability at least $2/3$. The M_v repetitions and a large enough C_M make the conditional branch reject with probability at least $5/6$. Reducing the two branch constants to allow primitive error proves the stated $2/3$. $\square$

Under the null, the deterministic query bound below is smaller than B_{alloc} when C_{alloc} is fixed large enough. Hence the ideal-target false-rejection probability of one entire amplified tester is at most $\delta/100$. Under the alternative, independent top-level repetitions and a Chernoff bound reduce constant error below $\delta/4$. The two approximation couplings in Section 6 use less than another $\delta/100$ each. Enlarging C_R handles the remaining slack and proves error at most δ .

5.2 The query recurrence

Let $Q(m, e)$ be the maximum hidden-query count of one core invocation, including anchors, with the master value $\mathcal{L}$ held fixed. The logarithms in Theorem 3.5 are at most $O(\mathcal{L})$ for every threshold that can arise. At a fixed face dimension s and signal level u , anchors plus raw-mean samples cost at most

$$C\mathcal{L}^4 \frac{u}{g} \max \left\{ 1, \frac{\sqrt{s}}{\kappa} \max\{u^{-2}, u^{-1}\} \right\}. \quad (19)$$

The dyadic sum must include $u > 1$. For $u \leq 1$, the dominant summand is $\sqrt{s}/(\kappa g u)$ and sums to $O(\sqrt{s}/(\kappa g^2))$; for $u > 1$, it is at most $\sqrt{s}/(\kappa g)$ per geometric scale and is smaller because $g \leq 1$. The anchor-only terms are also smaller. Thus

$$W_{\text{mean}}(s) \leq C \frac{\sqrt{s}}{\kappa g^2} \mathcal{L}^6. \quad (20)$$

Assume first that $\sigma m \geq 2$. Then

$$\frac{\sigma m}{2} \leq t \leq \sigma m, \quad s \leq \frac{3\sigma m}{2}, \quad g \geq \frac{\sigma e}{16\bar{\Lambda}}.$$

Using $\bar{\Lambda} = O(\kappa^{-5/2} \eta^{-4} \mathcal{L}^5)$ and $\sigma^{-3/2} = O(\mathcal{L}^{12})$ in (20) gives

$$W_{\text{mean}} \leq C \frac{\bar{\Lambda}^2 \sqrt{m}}{\kappa \sigma^{3/2} e^2} \mathcal{L}^6 \leq C \kappa^{-6} \eta^{-8} \frac{\sqrt{m}}{e^2} \mathcal{L}^{28}. \quad (21)$$

The recursive branch satisfies

$$W_{\text{rec}} \leq \sum_{v \in \text{Dyad}(e/4, 1)} C \frac{v}{e} \mathcal{L}^3 (1 + Q(t, v)). \quad (22)$$

Suppose inductively that

$$Q(t, v) \leq A \kappa^{-6} \eta^{-8} \sqrt{t} v^{-2} \mathcal{L}^{39}. \quad (23)$$

The dyadic identities $\sum_v v \leq 2$ and $\sum_v v^{-1} \leq 8/e$ imply that the recursive calls contribute at most

$$\frac{8CA}{\sqrt{C_\sigma}} \kappa^{-6} \eta^{-8} \frac{\sqrt{m}}{e^2} \mathcal{L}^{38}. \quad (24)$$

Choose the integer C_σ so that the coefficient in (24) is at most $A/2$; then choose A to dominate (21) and anchor terms.

The small-dimension regime requires a separate calculation. If $\sigma m < 2$, then $t = 1$ and $m < 2/\sigma = O(\mathcal{L}^8)$. Directly summing (19) gives

$$W_{\text{mean}} \leq C \kappa^{-6} \eta^{-8} e^{-2} \mathcal{L}^{32}. \quad (25)$$

Every recursive child is one-dimensional, and the total child cost is $O(\kappa^{-1} e^{-2} \mathcal{L}^4)$. We do not use the false identity $\sqrt{t} = \sqrt{\sigma} \sqrt{m}$ in this regime.

Strong induction now gives

$$Q(m, e) \leq A \kappa^{-6} \eta^{-8} \sqrt{m} e^{-2} \mathcal{L}^{39}. \quad (26)$$

The $O(\mathcal{L})$ root repetitions yield the exponent 40 in (6). Taking C_{alloc} larger than the resulting fixed coefficient proves that B_{alloc} is a deterministic upper bound on the number of fallible calls, so the earlier union bound is noncircular.

Combining Lemma 5.2, the recurrence, and the implementation coupling below proves Theorem 3.1.

6 VISIBLE-MODEL IMPLEMENTATION AND NUMERICAL ERROR

The statistical tester asks for independent samples from pinned target laws. These can be approximated from (J, h, ρ) without $Z_{J,h}$. On a supported face with r active spins, a random-scan heat-bath update chooses i uniformly and uses

$$Q^\rho(X_i = +1 \mid x_{-i}) = \frac{1}{1 + \exp[-2(h_i^\rho + \sum_{j \in V(\rho) \setminus \{i\}} J_{ij}x_j)]}. \quad (27)$$

The coupling in Lemma D.6 contracts Hamming distance by $1 - \kappa/r$ per update. Hence

$$T_{\text{mix}}(r, \zeta) = \left\lceil \frac{r}{\kappa} \log \frac{r}{\zeta} \right\rceil \quad (28)$$

updates from any state consistent with the pins give TV error at most ζ . The chain is restarted with fresh randomness for every visible sample.

Set

$$\zeta = \frac{\delta}{200B_{\text{alloc}}}. \quad (29)$$

Conditionally on each adaptive history, maximal coupling matches the next approximate target draw to an ideal independent draw except with probability at most ζ . Sequential composition therefore couples all visible draws in the adaptive transcript except on an event of probability at most $\delta/200$.

There is a second, distinct numerical error: a computer may approximate the logistic coin in (27). Define the deterministic update budget

$$T_{\text{alloc}} = B_{\text{alloc}} \left\lceil \frac{Cn}{\kappa} \log \frac{200nB_{\text{alloc}}}{\delta} \right\rceil. \quad (30)$$

Approximate each heat-bath probability to additive error at most $\delta/(200T_{\text{alloc}})$, clipped to $[0, 1]$. Coupling every approximate coin to its exact counterpart and taking a union bound changes the full transcript with probability at most another $\delta/200$. These two error budgets are separate.

Exact schedule arithmetic. Assume the supplied parameters and certificates are rational. The quantities $\ell_0, \mathcal{L}, \sigma, t, B_{\text{alloc}}$ and all probability budgets are then computed by exact integer and rational arithmetic. Although $\bar{\Lambda}$ and g contain $\kappa^{-5/2}$, every candidate u in the mean-test grid is either $2^j g$ or $\sqrt{s}$, and hence u^2 is rational. Positive quadratic-algebraic comparisons locating the cap $\sqrt{s}$ in a dyadic grid are decided exactly by squaring positive rational inequalities. Dyadic logarithm ceilings are found by integer bit length and comparison with powers of two. In particular $H = 4u^2$ and every tensor-test threshold are rational. Thus no floating-point comparison is used to decide a statistical branch.

Required primitives. The implementation uses only:

- exact uniform-subset sampling, independent random bits, and integer and rational arithmetic;
- the hidden SCND oracle, only on the supported faces certified in Proposition 4.1;
- exact comparison and ceiling of positive quadratic-algebraic schedule quantities by squared rational comparisons;
- stable arbitrary-precision evaluation of the logistic probability in (27); for a requested additive error α , fields outside the range $O(\log(1/\alpha))$ are handled by saturation, and standard range reduction evaluates $\exp$ inside that range;
- integer inner products $\langle W, W' \rangle$, repeated squaring of their integer powers, and rational threshold comparison by cross multiplication.

No partition function, exact target marginal, exact target sampler, or unproved anti-concentration primitive is assumed.

Running time and bit complexity. Let M denote the hidden-query bound in (6). A conservative real-arithmetic running-time bound is

$$O\left(nM^2 \log \log(2 + M^2) + \frac{Mn^2}{\kappa} \log \frac{200nB_{\text{alloc}}}{\delta}\right). \quad (31)$$

The first term evaluates all split inner-product statistics naively; the second recomputes a local field from scratch at every Glauber update. Maintaining all local fields changes the per-update factor n to the degree of the updated vertex. Including the visible input matrix, the memory bound is

$$O(\text{nnz}(J) + nM)$$

words, or $O(n^2 + nM)$ for a dense representation.

For rational input of bit length B_{in} , summing a local field and evaluating a logistic coin to the above accuracy requires precision polynomial in

$$B_{\text{in}} + \log n + \log(T_{\text{alloc}}/\delta).$$

At the largest tensor level, an exact integer $\langle W, W' \rangle^{2^{k_{\text{max}}}}$ has

$$O(2^{k_{\text{max}}} \log n + \log M) = O(\log M \log n) \quad (32)$$

bits, since $2^{k_{\text{max}}} = O(\log M)$. Threshold numerators and denominators have comparable polynomial length. Standard quasi-linear integer multiplication and arbitrary-precision exponential evaluation therefore make the implementation polynomial-time in n, M, B_{in} and the certificate precision. Equation (31) is a real-arithmetic count; multiplying by the corresponding precision cost gives a conservative bit-operation bound. No strongly polynomial bit-complexity claim is intended.

7 LITERATURE STATUS, ORACLE TAXONOMY, AND COMPARISON

The primary-source audit for this manuscript was completed on 20 July 2026. The theorem numbers below refer to CCKLW v2, BC v2, BCSV v3, AFL v2, CMT v3, Marton v1, and BCCPSV v1; the bibliography records the corresponding arXiv identifiers. We distinguish non-tolerant TV identity from tolerant estimation and from forward $D_{\text{KL}}(P||Q)$ throughout.

Oracle taxonomy. A General oracle returns an unconditional full sample. A Coordinate oracle fixes all but one coordinate and returns that remaining spin. A binary Subcube oracle fixes an arbitrary coordinate subset and returns all free coordinates; its all-star query is the General oracle. A Pairwise oracle conditions on an arbitrary two-point set $\{x, y\}$ in the whole domain, while unrestricted COND permits an arbitrary domain subset. Thus Pairwise and binary SCND are generally incomparable, and unrestricted COND is stronger than SCND. For binary cubes, the strong-alphabet product-set SUBCOND oracle of BC is equivalent to pin/free SCND. Prefix and marginal-prefix oracles respectively return a conditional suffix and only its next coordinate after fixing an initial segment. In particular, a Coordinate+General lower bound does not automatically lower-bound SCND; BCSV Definition 3.1 and Remark 3.3 make these implications explicit [4].

Zero-mass conventions also differ. BC and CCKLW define fallback answers, BCSV permits an arbitrary answer, and AFL returns an error. The present model obtains no information from an undefined conditional. Proposition 4.1 proves legality directly, so none of those fallback conventions is imported.

Uniformity and known-target TV testing. CCKLW prove non-tolerant TV uniformity testing under binary SCND with $\tilde{O}(\sqrt{n}/\varepsilon^2)$ queries [6, Theorems 1.1 and 2.1]. Their fixed-size localization, directed robust-Pisier inequality, and arbitrary-law mean tester are, respectively, Lemma 3.1, Theorem 3.2, and Theorems 1.3/4.3 [6]. Their target is uniform; none of these statements is an identity theorem for a biased product or dependent Ising target.

BC Theorem 4.2 gives non-tolerant known-target TV identity with $\tilde{O}(n^2/\varepsilon^2)$ strong-alphabet SUBCOND queries [2]. On the binary cube its conditioning sets are legitimate subcubes, but the theorem assumes target samples and exact one-dimensional target conditionals. Its target-anchored construction also invokes the paper’s uniform fallback when a hidden face has zero mass. Those visible primitives and that fallback are not available in our strict model.

AFL Theorem 5.8 and Lemma 5.4 give a balance-free, non-tolerant TV equivalence tester using prefix access to one distribution and marginal-prefix access to the other, with $\tilde{O}(n/\varepsilon^2)$ queries [1]. Orienting the hidden law on the prefix side makes hidden queries support-safe; approximate pinned Glauber samples can implement the full-support Ising target side. This supplies a useful linear-rate fallback, but not the square-root rate. AFL Theorem 6.1 and Lemmas 6.3–6.5 also prove the $\Omega(\sqrt{n}/\varepsilon^2)$ TV baseline against adaptive randomized SCND through an exact online query simulation [1].

KL identity and tolerant estimation. BCSV Theorems 4.1, 7.2, and 7.5 concern exact identity in forward KL, using Coordinate+General or SCND access and balance plus exact or FPRAS-visible conditional primitives [4]. Converting $d_{\text{TV}}(P, Q) \geq \varepsilon$ to $D_{\text{KL}}(P\|Q) \geq 2\varepsilon^2$ gives linear rather than square-root dependence on n . Their Theorem 7.9 is tolerant additive forward-KL estimation, not tolerant TV identity. Their Theorem 6.3 is an adaptive SCND lower bound for KL whose hard alternatives have only $O(\varepsilon/n)$ TV separation, whereas Theorem 6.9 is a TV lower bound only for Coordinate+General access. None is an adaptive-SCND TV obstruction at the parameter used here.

Approximate tensorization and spectral independence. CMT Theorem 2.1 and Corollary 2.3 establish high-temperature approximate tensorization of entropy under their small-interaction criterion [8]. Marton Theorem 2 and (1.7) give a Dobrushin forward-KL tensorization constant containing the reciprocal minimum one-site conditional mass [13]; it is therefore not field-uniform. BCCPSV Definitions 1.4–1.5 and Theorems 1.6, 1.7, and 1.13 connect spectral independence, mixing, and entropy factorization [3]. These KL/entropy inequalities do not themselves provide the TV relative random-restriction theorem used here, and we never recover our TV signal by Pinsker.

The realizable product benchmark. When both the visible and hidden laws are products, known-product identity uses $O(\sqrt{n}/\varepsilon^2)$ ordinary samples, uniformly over all biases, by CDKS Theorem 2 and independently the full-version Daskalakis–Pan Theorem 14 [7, 9]. Ordinary samples are all-star SCND calls. This realizable theorem is strictly weaker than Corollary 3.2, whose hidden law may have arbitrary dependence.

Table 1 separates the principal regimes. *Robust* means that the hidden law is arbitrary; *realizable* means that it is promised to belong to the indicated model class.

What is new and what is imported. The pinned/projected heat-bath robust-Pisier theorem (Theorem 3.3), the relative localization theorem (Theorem 3.4), and the dependent-null target-paired mean test (Theorem 3.5) are proved here. The external proof ingredients used without reproving their general forms are precisely the finite Dobrushin comparison theorem [10, Theorem 8.20], the Dobrushin martingale MGF bound [12, Theorem 4.2 and Section 5.1, Corollary 2], path coupling [5], and Hoeffding’s bounded-summand tail bound [11, Theorem 2]. Appendix D specializes

Table 1. Identity testing with binary SCND. The $\tilde{\Theta}$ entries compare only the n, ε dependence; the κ, η exponents are not claimed optimal.

Visible target	Hidden law	Balance	TV query status	Comment
Uniform	arbitrary	$\eta = 1/2$	$\tilde{\Theta}(\sqrt{n}/\varepsilon^2)$	CCKLW upper; Theorem 3.6 gives the adaptive-SCND lower baseline.
Arbitrary product	arbitrary	finite-target η	$O(\eta^{-8}\sqrt{n}\varepsilon^{-2})$ polylog	Corollary 3.2; no bias-uniform robust bound is claimed.
Arbitrary product	product (realizable)	none	$\Theta(\sqrt{n}/\varepsilon^2)$	Ordinary-sample product identity [7, Theorem 2] [9, Theorem 14]; uniform over biases. Theorem 3.6 supplies the class lower bound.
Balanced Dobrushin Ising	arbitrary	promised valid $\eta > 0$	$\tilde{\Theta}(\sqrt{n}/\varepsilon^2)$	Theorem 3.1; robust, with explicit $\kappa^{-6}\eta^{-8}$ upper dependence.
Balanced Dobrushin Ising	Dobrushin Ising (realizable)	promised valid $\eta > 0$	$\tilde{\Theta}(\sqrt{n}/\varepsilon^2)$	Upper bound specializes from the robust theorem; the balanced product subfamily gives the lower bound.
Arbitrary finite-field Dobrushin Ising	arbitrary	$\eta(H_*) > 0$ may be tiny	the theorem applies, but not field-uniformly	Proposition 2.3 gives an efficiently checkable conservative certificate; dependence can be exponential in the field magnitude.

each one in the exact normalization used. All other analytic, restriction, tensor-growth, recursion, oracle-legality, and adaptive lower-bound lemmas are proved in this manuscript.

A FIELD-UNIFORM HEAT-BATH ROBUST PISIER

This appendix proves Theorem 3.3. We first transfer the uniform-cube theorem of CCKLW to arbitrary product biases, then treat a small absolute-interaction regime by a Gaussian product mixture, reduce the full Dobrushin regime to that base case, and finally prove projection closure.

A.1 Arbitrary product fibres

LEMMA A.1 (DIRECTED PRODUCT ROBUST PISIER). *Let $\mu = \bigotimes_{i=1}^m \mu_i$ be an arbitrary binary product law. Independently draw $X \sim \mu$ and resampling values $X'_i \sim \mu_i$, and let $X^{(i)} = (X_{-i}, X'_i)$. For every orientation of the coordinate fibres and every centered F ,*

$$\mathbb{E}|F(X)| \leq C \log(m+1) \mathbb{E} \left| \sum_i \epsilon_i \mathbf{1}\{X \rightarrow X^{(i)}\} (F(X) - F(X^{(i)})) \right|. \quad (33)$$

The constant is uniform over the biases μ_i .

PROOF. Draw independent $U, V \sim \mu$ and condition on the coordinatewise unordered pairs $\mathcal{B}_i = \{U_i, V_i\}$. On coordinates where $U_i \neq V_i$, the choice of which endpoint belongs to U is an independent fair bit; the remaining coordinates are fixed. Conditional on $\mathcal{B}$, U is therefore the image of a uniform point on a cube of dimension at most m , and flipping the i th fair bit replaces U_i by V_i . Apply CCKLW's directed robust-Pisier theorem [6, Theorem 3.2] to $F(U) - \mathbb{E}[F(U) \mid \mathcal{B}]$. Finally,

$$\mathbb{E}|F(U) - \mathbb{E}F(U)| \leq \mathbb{E}|F(U) - F(V)| \leq 2\mathbb{E}|F(U) - \mathbb{E}(F(U) \mid \mathcal{B})|.$$

Averaging over $\mathcal{B}$ proves (33). Self-loops contribute zero. $\square$

A.2 The small-interaction Gaussian block

For an Ising interaction J put $s_J = \max_i \sum_{j \neq i} |J_{ij}|$. Assume first that $2s_J < 1$ and set

$$A = J + s_J I. \quad (34)$$

Gershgorin's theorem gives $A \geq 0$ and $\max_i \sum_j |A_{ji}| \leq 2s_J$. The diagonal shift changes the Ising Hamiltonian by the constant $s_J m/2$. The (possibly degenerate) Hubbard–Stratonovich identity gives a joint law

$$d\nu(x, g) \propto e^{h^T x + g^T x} dN(0, A)(g) \quad (35)$$

with conditionals

$$X \mid G = g = \bigotimes_i \frac{e^{(h_i + g_i)x_i}}{2 \cosh(h_i + g_i)}, \quad G \mid X = x = N(Ax, A). \quad (36)$$

Let P_G be the two-block kernel $X \rightarrow G \rightarrow Y$. It is reversible with respect to Q and positive semidefinite, being T^*T for the conditional expectation operator T . Couple neighboring inputs x, x^i with the same Gaussian noise and common uniforms at the output. The derivative of $a \mapsto (1 + e^{-2a})^{-1}$ is at most $1/2$, hence

$$\mathbb{E} d_H(Y_x, Y_{x^i}) \leq \frac{1}{2} \sum_j |(A(x - x^i))_j| = \sum_j |A_{ji}| \leq 2s_J. \quad (37)$$

Path coupling contracts the oscillation seminorm by $2s_J$. Applying this to a nonconstant eigenfunction, and using positive semidefiniteness, shows

$$\text{gap}(P_G) \geq 1 - 2s_J. \quad (38)$$

The finite coarea lemma, Lemma D.3, followed by conditional iid-ness of two X samples given G , yields

$$\mathbb{E}_Q |F - \mathbb{E}_Q F| \leq \frac{8}{1 - 2s_J} \mathbb{E}_\nu |F(X) - \mathbb{E}(F(X) \mid G)|. \quad (39)$$

Conditional on G , Lemma A.1 applies with a constant $C \log(m + 1)$.

It remains to compare the resulting product-fibre flips with true Ising heat-bath flips. At x define

$$t_i = x_i \left(h_i + \sum_{j \neq i} J_{ij} x_j \right), \quad k(t) = \frac{1}{1 + e^{2t}}.$$

The true flip probability is $k(t_i)$. Under the latent fibre, conditioned back on $X = x$, the marginal flip probability equals

$$p_i(x) = \mathbb{E}_Z k(t_i + s_J + \sqrt{s_J} Z), \quad Z \sim N(0, 1). \quad (40)$$

If $t_i \geq 0$, then $k(u) \leq e^{-2u}$ and $\mathbb{E} e^{-2\sqrt{s_J} Z} = e^{2s_J}$, so $p_i(x) \leq e^{-2t_i} \leq 2k(t_i)$. If $t_i < 0$, the same conclusion follows from $p_i(x) \leq 1 \leq 2k(t_i)$. After G is averaged out, the latent flip indicators may be correlated, but Lemma D.2 with factor two, followed by the upper and lower Khintchine bounds in Lemma D.1, converts them to independent true heat-bath indicators. We have proved the small-interaction estimate

$$K_m \leq \frac{C \log(m + 1)}{1 - 2s_J}. \quad (41)$$

No bound on h was used.

A.3 Defective blocks and the full Dobrushin range

Write $c = 1 - \kappa$. For every integer $L \geq 1$ there is a partition $V = B_1 \sqcup \cdots \sqcup B_L$ such that

$$\sum_{j \in B_\ell} C_{ij} \leq c/L \quad \text{for every } i \in B_\ell. \quad (42)$$

Indeed, choose an L -coloring minimizing the total weight of monochromatic undirected edges. Moving a vertex to any color cannot lower that weight; its weight to its present color is therefore no larger than the average weight to all colors, at most c/L .

Put $\delta = c/L$ and consider random-block heat bath, which chooses one block uniformly and resamples it conditional on its complement. Couple two boundaries differing at one site j . If the chosen block contains j , the coupled outputs agree. Otherwise Dobrushin comparison bounds the new disagreement vector in block B by $(I - C_{BB})^{-1}C_{B,j}$, of ℓ_1 norm at most $(1 - \delta)^{-1} \sum_{i \in B} C_{ij}$. Summing over blocks gives

$$\mathbb{E}d_H(X_1, Y_1) \leq 1 - \frac{1}{L} \frac{\kappa - \delta}{1 - \delta}. \quad (43)$$

The block operator is an average of orthogonal conditional-expectation projections, hence is reversible and positive semidefinite. The same oscillation argument used above gives gap at least γ_B/L , where

$$\gamma_B = \frac{\kappa - \delta}{1 - \delta}. \quad (44)$$

Apply Lemma D.3 to block dynamics. For block B_ℓ put

$$D_\ell = \mathbb{E} |F - \mathbb{E}(F \mid X_{B_\ell^c})|,$$

and let S_ℓ be the directed single-site heat-bath score restricted to B_ℓ . Conditional one-site heat-bath flips within a block are the original heat-bath flips. Since the normalized block kernel has gap γ_B/L , coarea, conditional iid resampling, and the within-block robust inequality give

$$\begin{aligned} \mathbb{E}|F - \mathbb{E}F| &\leq \frac{8L}{\gamma_B} \mathbb{E}_\ell D_\ell = \frac{8}{\gamma_B} \sum_{\ell=1}^L D_\ell \\ &\leq \frac{8R(\delta, m)}{\gamma_B} \sum_{\ell=1}^L \mathbb{E}|S_\ell|. \end{aligned}$$

Use common independent signs and heat-bath indicators to write $S = \sum_\ell S_\ell$. Conditional on all unsigned coefficients, Lemma D.1 and Cauchy–Schwarz give

$$\sum_{\ell=1}^L \mathbb{E}_\epsilon |S_\ell| \leq \sum_{\ell=1}^L \|a_{B_\ell}\|_2 \leq \sqrt{L} \|a\|_2 \leq \sqrt{3L} \mathbb{E}_\epsilon |S|.$$

Thus, if $R(c, m)$ denotes the worst constant at influence row sum c ,

$$R(c, m) \leq \frac{8\sqrt{3L}}{\gamma_B} \max_{r \leq m} R(\delta, r). \quad (45)$$

Choose

$$d_* := \min\{\tanh(1/4), \kappa/2\}, \quad L := \max\{1, \lceil (1 - \kappa)/d_* \rceil\}. \quad (46)$$

Then $\delta \leq d_*$ and $\gamma_B \geq \kappa/2$. Moreover, within every block B_ℓ ,

$$s_{J,\ell} := \max_{i \in B_\ell} \sum_{j \in B_\ell \setminus \{i\}} |J_{ij}| = \max_{i \in B_\ell} \sum_{j \in B_\ell \setminus \{i\}} \operatorname{arctanh}(C_{ij}) \leq \operatorname{arctanh}(d_*) \leq 1/4.$$

Thus (41) applies within each block with $1 - 2s_{J,\ell} \geq 1/2$. Substitution into (45) proves (8) for every pin, since pinning only deletes influences.

A.4 Closure under coordinate projection

Let μ be a strictly positive full law satisfying (7) with constant K , and let $\nu = \mu_R$ be a coordinate projection. Lift a function g on R to $G(x) = g(x_R)$ and lift its edge orientation. For a projected state z and direction $i \in R$, write, over outside states u ,

$$a_u = \mu(z, u), \quad b_u = \mu(z^i, u), \quad A_0 = \sum_u a_u = \nu(z), \quad A_1 = \sum_u b_u = \nu(z^i).$$

The stationary full heat-bath flow, averaged conditional on $X_R = z$, is

$$A_0 p_i(z) = \sum_u \frac{a_u b_u}{a_u + b_u}.$$

With $c_u = a_u + b_u$ and $\theta_u = a_u/c_u$, concavity of $\theta(1 - \theta)$ gives

$$\sum_u \frac{a_u b_u}{a_u + b_u} \leq \frac{A_0 A_1}{A_0 + A_1}. \quad (47)$$

Thus $p_i(z)$ is at most the projected heat-bath flip probability $r_i(z) = A_1/(A_0 + A_1)$.

After outside variables are averaged out, the full-model flip indicators can be correlated, but their marginals are at most $r_i(z)$. Apply Lemma D.2 with factor one and then Lemma D.1. The projected score loses at most $\sqrt{3}/(1 - e^{-1})$. Therefore every projection satisfies (7) with a universal multiple of K , completing the proof of Theorem 3.3.

B THE BALANCED RELATIVE-RESTRICTION THEOREM

We prove Theorem 3.4 in an abstract form. Let Q be strictly positive and hereditarily η -balanced. Assume every pinned projection satisfies (7) with constant at most K , and assume the influence radius (9) is at most V . We prove (11) with the parameters in (10); Theorem 3.3 and the comparison bound at the end of this appendix then imply (13).

B.1 Relative edges

Fix a projection R , write $p = P_R$, $q = Q_R$, and $f = p/q$. For an undirected nontrivial edge $e = \{x, x^i\}$ with $p(x) + p(x^i) > 0$, put

$$a_e = \frac{q(x)}{q(x) + q(x^i)}, \quad b_e = \frac{p(x)}{p(x) + p(x^i)}, \quad u_e = |b_e - a_e|, \quad (48)$$

and

$$w_e = \frac{|f(x) - f(x^i)|}{\max\{f(x), f(x^i)\}}. \quad (49)$$

Edges with zero hidden mass can be discarded. Direct algebra gives

$$u_e = w_e \max\{b_e(1 - a_e), a_e(1 - b_e)\}. \quad (50)$$

The maximum as a function of b_e is minimized at $b_e = a_e$. Hereditary balance therefore implies

$$c_\eta w_e \leq u_e \leq w_e, \quad c_\eta = \eta(1 - \eta). \quad (51)$$

Call an edge *high* if its endpoint likelihood ratio is at least $L_\eta = 3(1 - \eta)/\eta$, and orient it from higher to lower likelihood. All other nonzero edges are *moderate*. Bucket a moderate edge into the unique scale $s \geq 1$ such that

$$2^{-s} < u_e \leq 2^{-s+1}. \quad (52)$$

Within each moderate bucket, repeatedly delete a vertex of maximum current degree and orient all its remaining incident edges outward. This greedy orientation has the following property: if vertices in a set A have out-degree at most g , then at most g edges are oriented from A into any fixed vertex outside A . Indeed, consider the first source in A whose edge is oriented into a fixed target. At that moment the source has maximum current degree, its current degree equals its eventual out-degree and is at most g , and every later incoming edge is among the target's then-current incident edges. Thus there are at most g such edges.

The endpoint ratios used below are

$$p(x) \geq 3p(x^i) \quad \text{on a high edge } x \rightarrow x^i, \quad (53)$$

and

$$\max\{p(x)/p(x^i), p(x^i)/p(x)\} < B_\eta = 3 \left(\frac{1-\eta}{\eta} \right)^2 \quad (54)$$

on a moderate edge. To see this, the target endpoint ratio and its reciprocal are at most $(1-\eta)/\eta$. Multiply by the likelihood ratio and use the definitions of L_η and B_η .

B.2 Projection TV supplies directed degree

Let $\beta = d_{\text{TV}}(p, q)$. Apply (7) to $F = f$ and the union of the orientations just defined. Since $\mathbb{E}_q|f-1| = 2\beta$, upper Khintchine and the change of measure $q(x)f(x) = p(x)$ give

$$\frac{2\beta}{K} \leq \mathbb{E}_{X \sim p, I} \sqrt{\sum_{i: X \rightarrow X^i} I_i \left(\frac{f(X) - f(X^i)}{f(X)} \right)^2}. \quad (55)$$

There is no 0/0: a zero-to-positive likelihood edge is high and oriented from its positive endpoint. Dropping the activations increases the right side. A high outgoing coefficient is $w_e \leq 1$. On a moderate outgoing edge at scale s , (51) and the likelihood-ratio bound give

$$\frac{|f(x) - f(x^i)|}{f(x)} \leq \frac{2L_\eta}{c_\eta} 2^{-s}.$$

If $d_{\text{hi}}(x)$ and $d_s(x)$ are the corresponding outgoing degrees, then

$$\frac{2\beta}{K} \leq \mathbb{E}_p \sqrt{d_{\text{hi}}(X)} + \frac{2L_\eta}{c_\eta} \sum_{s \geq 1} 2^{-s} \mathbb{E}_p \sqrt{d_s(X)}. \quad (56)$$

All degrees are at most m . The tail after

$$S_\beta = 2 + \left\lceil \log_2 \frac{16KL_\eta \sqrt{m}}{c_\eta \beta} \right\rceil \quad (57)$$

is at most $\beta/(2K)$. Consequently either

$$\mathbb{E}_p \sqrt{d_{\text{hi}}(X)} \geq \frac{\beta}{4K}, \quad (58)$$

or for some $s \leq S_\beta$,

$$\mathbb{E}_p \sqrt{d_s(X)} \geq \frac{c_\eta 2^s \beta}{8KL_\eta S_\beta}. \quad (59)$$

B.3 Edges are parent-face residuals

Choose a $(t + 1)$ -set S , an anchor y , and $i \in S$. The projected edge on $(S \setminus \{i\})^c$ at the anchor has

$$a_e = Q(B_i = y_i \mid X_{S^c} = y_{S^c}), \quad b_e = P(B_i = y_i \mid X_{S^c} = y_{S^c}).$$

It follows that

$$u_e = |\Delta(\rho(S, y))_i|. \quad (60)$$

A high outgoing edge has

$$|\Delta(\rho(S, y))_i| \geq c_\eta(1 - L_\eta^{-1}) \geq 2c_\eta/3, \quad (61)$$

while a scale- s moderate edge has $|\Delta(\rho(S, y))_i| > 2^{-s}$.

B.4 High likelihood edges

LEMMA B.1 (HIGH-EDGE PAIR). *Suppose the i - and j -projection edges at the same parent anchor are both high and outgoing. Let ρ_i be the child obtained by pinning i to the anchor and define ρ_j analogously. Then*

$$\max\{|\Delta(\rho_i)_j|, |\Delta(\rho_j)_i|\} \geq c_\eta/10. \quad (62)$$

PROOF. Marginalize both laws to (i, j) conditional on the outer pins and let f_{ab} be the four cell likelihood ratios, with 11 the anchor cell. If both residuals in (62) were below $c_\eta/10$, then (51) would make the two adjacent likelihood-edge weights below $1/10$. Hence, with $D_0 = 10/9$,

$$D_0^{-1}f_{11} < f_{10}, f_{01} < D_0f_{11}. \quad (63)$$

Only these two adjacent ratios are controlled; f_{00} is neither controlled nor needed. The likelihood of the anchor i -marginal is at most D_0f_{11} . In the opposite i -marginal the target conditional assigns mass at least η to the 01 cell, so that marginal likelihood is at least $\eta f_{11}/D_0$. Their ratio is at most

$$\frac{D_0^2}{\eta} = \frac{100}{81\eta} < 3 \frac{1 - \eta}{\eta} = L_\eta,$$

contradicting that the i -edge is high outgoing. $\square$

Fix a selected collection of projection vertices whose high out-degree is in $[d, 2d]$, and let λ be the joint probability of this selection under a uniform t -projection and hidden anchor. Call a t -face good if

$$\|\Delta\|_2 \geq h_{hi} := c_\eta \sqrt{d}/10, \quad (64)$$

and let G be the $\mathcal{D}_P(t)$ probability of a good face.

At a fixed parent let X_i indicate that the distinguished i -incidence is selected and high outgoing and its anchor child is not good. Put $M = \sum_{i \in S} X_i$. Apply Lemma B.1 to every pair of counted indices. Every pair contributes a distinct coordinate of squared size at least $(c_\eta/10)^2$ to one of its two children. Since every counted child has squared norm below $(c_\eta/10)^2 d$,

$$\binom{M}{2} (c_\eta/10)^2 < M(c_\eta/10)^2 d,$$

so

$$M \leq 2d + 1 \leq 3d. \quad (65)$$

We display the adjacent-level normalization. Generate an incidence by choosing a uniform t -set, a hidden anchor, and then a uniform additional coordinate among the $m - t$ pinned coordinates. Equivalently, choose a uniform $(t + 1)$ -set S , a uniform distinguished $i \in S$, and an anchor. Thus

$$\mathbb{E}_{\text{parent}} \sum_{i \in S} X_i = (t + 1) \mathbb{P}(X_i = 1). \quad (66)$$

For a selected source there are at least d outgoing choices, and discarding good anchor children costs at most G ; hence

$$\mathbb{P}(X_i = 1) \geq (\lambda - G) \frac{d}{m - t}. \quad (67)$$

If $G \geq \lambda/2$, the t -face expectation is at least $\lambda c_\eta \sqrt{d}/20$. Otherwise (66)–(65) and $\sqrt{M} \geq M/\sqrt{3d}$ imply

$$\begin{aligned} \mathbb{E}_{\mathcal{D}_P(t+1)} \|\Delta\|_2 &\geq (2c_\eta/3) \mathbb{E} \sqrt{M} \\ &\geq cc_\eta \lambda \sqrt{d} \frac{t+1}{m-t} \geq cc_\eta \lambda \sqrt{d} \frac{t}{m}. \end{aligned} \quad (68)$$

Both branches yield

$$\mathbb{E}_{\mathcal{D}_P(t)} \|\Delta\|_2 + \mathbb{E}_{\mathcal{D}_P(t+1)} \|\Delta\|_2 \geq cc_\eta \lambda \frac{t}{m} \sqrt{d}. \quad (69)$$

B.5 Moderate edges

Fix a scale s and selected vertices of out-degree in $[d, 2d]$, with joint probability λ . Put

$$D = 8(1 + V), \quad \Gamma = D^2 d, \quad h_s = 2^{-s} D \sqrt{d}/4, \quad (70)$$

and call a t -face good if $\|\Delta\|_2 \geq h_s$. Let G denote its probability.

At a parent and pivot i , write $a = Q(B_i = 1 \mid \rho)$, $b = P(B_i = 1 \mid \rho)$ and let Δ^+, Δ^- be the residual vectors on the two i -children. With

$$v^{\rho, i} = (Q(B_j = 1 \mid \rho, B_i = 1) - Q(B_j = 1 \mid \rho, B_i = 0))_{j \in V(\rho) \setminus \{i\}},$$

the exact mixture identity is

$$\Delta_{-i} = b\Delta^+ + (1 - b)\Delta^- + (b - a)v^{\rho, i}. \quad (71)$$

If both children are not good and the i -edge has scale s , then

$$\|\Delta_{-i}\|_2 < h_s + 2^{-s+1}V. \quad (72)$$

Let X_i count selected scale- s outgoing incidences for which both i -children are not good. If $M = \sum_i X_i > 0$, choose one counted pivot. Every other counted coordinate has magnitude greater than 2^{-s} by (60). Equations (70)–(72) imply

$$2^{-s} \sqrt{M - 1} \leq h_s + 2^{-s+1}V \leq (D/2)2^{-s} \sqrt{d},$$

and therefore

$$M \leq D^2 d = \Gamma. \quad (73)$$

It remains to account for edges whose other child is good. Before that exclusion, the incidence probability is at least $(\lambda - G)d/(m - t)$. Under the maximum-current-degree orientation, any fixed good target receives at most $2d$ edges from degree- $[d, 2d]$ sources. Changing from source mass to target mass costs at most B_η by (54). Hence the excluded probability is at most $2B_\eta dG/(m - t)$, and

$$\mathbb{P}(X_i = 1) \geq [\lambda - (1 + 2B_\eta)G] \frac{d}{m - t}. \quad (74)$$

If $G \geq \lambda/[2(1 + 2B_\eta)]$, good t -faces already contribute

$$\frac{\lambda 2^{-s} D \sqrt{d}}{8(1 + 2B_\eta)}.$$

Otherwise (74), the incidence identity (66), the cap (73), and $\sqrt{M} \geq M/(D\sqrt{d})$ give the $(t+1)$ -face contribution. In both cases,

$$\mathbb{E}_{\mathcal{D}_P(t)} \|\Delta\|_2 + \mathbb{E}_{\mathcal{D}_P(t+1)} \|\Delta\|_2 \geq c \frac{t}{m} \frac{\lambda 2^{-s} \sqrt{d}}{B_\eta D}. \quad (75)$$

B.6 Bucketing and completion

For each uniform t -set T , let $\beta_T = d_{TV}(P_{T^c}, Q_{T^c})$. Values below $\alpha_t/2$ contribute at most $\alpha_t/2$ to the expectation. Among the $H_{\alpha_t} = O(\log(2/\alpha_t))$ dyadic ranges between $\alpha_t/2$ and one, choose $[\beta_0, 2\beta_0]$ with probability ζ such that

$$\zeta \beta_0 \geq \frac{\alpha_t}{4H_{\alpha_t}}. \quad (76)$$

Apply (58)–(59) to every selected T . Pigeonhole over high versus at most R_{α_t} moderate scales, then over the at most $\lceil \log_2 m \rceil + 1$ degree ranges $[d, 2d)$. There is a common type and degree with joint probability λ such that, in the high case,

$$\lambda \sqrt{d} \geq \frac{c \zeta' \beta_0}{K \log(2m)}, \quad (77)$$

and, in a moderate scale s ,

$$\lambda \sqrt{d} \geq \frac{c \zeta' c_\eta 2^s \beta_0}{KL_\eta R_{\alpha_t} \log(2m)}, \quad (78)$$

where $\zeta' \geq \zeta/(R_{\alpha_t} + 1)$. These inequalities follow by summing $\sqrt{2d}\mathbb{P}(d \leq D < 2d)$ over degree buckets; no pointwise bucket probability is assumed.

Insert (77) into (69), or (78) into (75); in the latter, 2^s cancels 2^{-s} . Then use (76). The moderate bound is the smaller and gives exactly (11).

Finally, for a pinned Ising target, maximal-coupling iteration gives the coordinatewise Dobrushin comparison

$$d \leq (I - C_{-i,-i})^{-1} C_{-i,i}.$$

Therefore

$$V \leq \|d\|_1 \leq \|(I - C_{-i,-i})^{-1}\|_1 \|C_{-i,i}\|_1 \leq \frac{1 - \kappa}{\kappa}, \quad (79)$$

so $D_V \leq 8/\kappa$. Theorem 3.3 gives $K = O(\kappa^{-3/2} \log(m+1))$ also for projections. Since

$$\frac{c_\eta}{L_\eta B_\eta} = \frac{\eta^4}{9(1-\eta)^2},$$

and each of $H_a, R_a, \log(2m)$ is bounded by a logarithm in (13), simplifying (11) proves Theorem 3.4.

C THE TARGET-PAIRED TENSOR MEAN TEST

We prove Theorem 3.5 with explicit tensor thresholds. The argument is inspired by CCKLW's tensor blow-up architecture [6, Lemma 4.1 and Theorem 4.3], but the paired dependent null and all moment bounds are proved here. Since every bit-mean residual has norm at most $\sqrt{m}$, the test returns ACCEPT immediately when $u > \sqrt{m}$. Henceforth assume $0 < u \leq \sqrt{m}$ and set $H = 4u^2 \leq 4m$.

C.1 Pairing and tensor identities

Let $X \sim P^\rho$ and $Y \sim Q^\rho$ be independent spins and put

$$W = (X - Y)/2 \in \{-1, 0, 1\}^m. \quad (80)$$

In bit notation, $\mathbb{E}W = \Delta(\rho)$. Under the null, W is centered. Lemma D.5 applied independently to X and Y gives

$$\log \mathbb{E} e^{\lambda \langle u, W \rangle} \leq \frac{\tau^2 \lambda^2 \|u\|_2^2}{2}, \quad \tau = 1/\kappa. \quad (81)$$

For $k \geq 0$, let $s_k = 2^k$ and

$$V_k(W) = W^{\otimes s_k}, \quad v_k = \mathbb{E}V_k, \quad r_k = \|v_k\|_2^2.$$

If $\Gamma_k = \mathbb{E}[V_k V_k^\top]$, vectorization gives the exact identity

$$r_{k+1} = \|\Gamma_k\|_F^2. \quad (82)$$

Draw two independent batches $W_1, \dots, W_q$ and $W'_1, \dots, W'_q$ and define

$$Z_k = \left\langle \frac{1}{q} \sum_{a=1}^q V_k(W_a), \frac{1}{q} \sum_{b=1}^q V_k(W'_b) \right\rangle = \frac{1}{q^2} \sum_{a,b=1}^q \langle W_a, W'_b \rangle^{s_k}. \quad (83)$$

Thus no tensor is materialized.

LEMMA C.1 (EXACT SPLIT-STATISTIC VARIANCE). *For every law of W ,*

$$\mathbb{E}Z_k = r_k, \quad \text{Var } Z_k \leq \frac{r_{k+1}}{q^2} + \frac{2r_k \sqrt{r_{k+1}}}{q}. \quad (84)$$

PROOF. Let $\Sigma_k = \Gamma_k - v_k v_k^\top$. Independence of the two empirical means gives

$$\text{Var } Z_k = \frac{\text{tr}(\Sigma_k^2)}{q^2} + \frac{2v_k^\top \Sigma_k v_k}{q}.$$

Since $\Gamma_k \geq v_k v_k^\top$, $\text{tr}(\Sigma_k^2) \leq \text{tr}(\Gamma_k^2)$ and $v_k^\top \Sigma_k v_k \leq r_k \|\Gamma_k\|_F$. Use (82). $\square$

C.2 Growth after a failed level

Fix an energy parameter $H > 0$ and set

$$t_0 = H/8, \quad t_{k+1} = q^2 t_k^2 / 1000. \quad (85)$$

If $r_k \geq 2t_k$ but $\mathbb{P}(Z_k > t_k) < 2/3$, then $\mathbb{P}(|Z_k - r_k| \geq r_k/2) > 1/3$. By Chebyshev and (84), with $z = \sqrt{r_{k+1}}/(qr_k)$,

$$\frac{1}{3} < 4z^2 + 8z.$$

Since $4(1/30)^2 + 8/30 < 1/3$, $z > 1/30$, and hence

$$r_{k+1} > q^2 r_k^2 / 900 \geq 2t_{k+1}. \quad (86)$$

This conclusion is distribution-free; large alternative variance becomes the next tensor signal.

C.3 Null moments and explicit arithmetic

If a centered scalar A is subgaussian with proxy $\tau^2 \|u\|_2^2$, tail integration gives, for $p \geq 2$,

$$\mathbb{E}|\langle u, W \rangle|^p \leq (4\tau^2 p \|u\|_2^2)^{p/2}. \quad (87)$$

For independent null samples W, W' , condition on W' and use $\|W'\|_2^2 \leq m$. For even $s = s_k$,

$$r_k \leq (4\tau^2 s m)^{s/2}, \quad \sqrt{r_{k+1}} \leq (8\tau^2 s m)^{s/2} =: B_s. \quad (88)$$

We now fix the universal integer

$$D_0 = 2^{30} \quad (89)$$

and choose

$$q = \left\lceil D_0 \tau \sqrt{m} \max\{H^{-1}, H^{-1/2}\} \right\rceil. \quad (90)$$

The closed form of (85) is

$$t_k = (q^2/1000)^{s-1} (H/8)^s, \quad s = 2^k. \quad (91)$$

LEMMA C.2 (THRESHOLD DOMINATION). *With (89)–(90), for every power of two $2 \leq s \leq 8 \log_2(2 + q^2)$,*

$$t_k \geq 4B_s. \quad (92)$$

PROOF. Put $c = 1/1000$ and $z = \tau \sqrt{m} \max\{1, H^{-1}\} \geq 1$. Directly from (91),

$$\frac{t_k}{B_s} = \frac{1}{cq^2} \left(\frac{cq^2 H}{16\sqrt{2} \tau \sqrt{sm}} \right)^s. \quad (93)$$

At $s = 2$, the ratio is at least $cD_0^2/1024 > 4$. Furthermore $q \leq 2D_0z$ and the parenthesized base at $s = 2$ is at least $cD_0^2z/32$. For $D_0 = 2^{30}$,

$$\frac{cD_0^2z}{32} \geq 8 \log_2(2 + 4D_0^2z^2) \quad (z \geq 1).$$

The difference has positive derivative for $z \geq 1$, and the inequality at $z = 1$ is immediate from $D_0^2 = 2^{60}$. At a general power of two s , the base is its $s = 2$ value times $\sqrt{2/s}$; dividing (93) by its value at $s = 2$ gives at least $A^{s-2}(2/s)^{s/2}$ with $A \geq s$. This is at least one for every $s \geq 4$, proving the claim. $\square$

Under the null, Lemma C.2, (88), and Lemma C.1 give, for $k \geq 1$, $\mathbb{E}Z_k \leq t_k/4$ and

$$\mathbb{P}(Z_k > t_k) \leq (q^{-2} + 2q^{-1})/9.$$

At $k = 0$, $\text{Var } Z_0 = r_1/q^2 \leq 8\tau^2 m/q^2$, so $\mathbb{P}(Z_0 > t_0) \leq 512D_0^{-2}$. Since $H \leq 4m$ and $\tau \geq 1$, (90) gives $q \geq D_0/2$. The definition of K below then gives

$$512D_0^{-2} + (K+1)(q^{-2} + 2q^{-1})/9 < 1/12.$$

Thus, in one null repetition, the event that any tested level marks has probability below $1/12$.

C.4 Algorithm and proof

Let K be the least integer such that

$$S := 2^{K+1} \geq 8 \log_2(2 + q^2). \quad (94)$$

In one repetition, draw the two batches in (83), compute every Z_k for $0 \leq k \leq K$, and mark the repetition if at least one level satisfies $Z_k > t_k$. Repeat

$$R = 72 \left\lceil \log_2 \frac{2}{\xi} \right\rceil \quad (95)$$

times and reject if a strict majority of repetitions are marked. The preceding $1/12$ null bound, Hoeffding's inequality, and the choice of R give completeness error at most $\xi/2$.

For soundness, recall that $H = 4u^2$. If $\|\Delta\|_2 \geq u$, then $r_0 = \|\mathbb{E}W\|_2^2 \geq H/4 = 2t_0$. If every level marked with probability below $2/3$, repeated application of (86) would give $r_{K+1} \geq 2t_{K+1}$. But V_{K+1} has m^S coordinates of magnitude at most one, so $r_{K+1} \leq m^S$. On the other hand,

$$\frac{t_{K+1}}{m^S} = \frac{1000}{q^2} \left(\frac{q^2 H}{8000m} \right)^S > 1. \quad (96)$$

Indeed, (90) and $D_0 = 2^{30}$ make the parenthesized base at least four in both regimes $H \leq 1$ and $H > 1$, and (94) dominates the prefactor. This contradiction shows that some fixed level marks with probability at least $2/3$, so a repetition is marked with probability at least $2/3$. The majority in (95) gives soundness error at most $\xi/2$.

Each repetition uses $2q$ hidden and $2q$ visible samples. Substituting $H = 4u^2$ into (90) and using (95) yields (14). Evaluating (83) takes $O(RKq^2m)$ arithmetic operations and $O(qm)$ words of memory. This completes the proof of Theorem 3.5.

D FINITE FUNCTIONAL AND COUPLING LEMMAS

For completeness, this appendix fixes the normalizations of the standard finite inequalities used above.

LEMMA D.1 (KHINTCHINE WITH UNIVERSAL CONSTANTS). *For independent Rademacher signs ϵ_i and real z_i ,*

$$\frac{1}{\sqrt{3}} \|z\|_2 \leq \mathbb{E} \left| \sum_i \epsilon_i z_i \right| \leq \|z\|_2.$$

PROOF. For $S = \sum_i \epsilon_i z_i$, $\mathbb{E}S^2 = \|z\|_2^2$ and $\mathbb{E}S^4 \leq 3 \|z\|_2^4$. Hölder gives $\mathbb{E}|S| \geq (\mathbb{E}S^2)^{3/2}/(\mathbb{E}S^4)^{1/2}$; Jensen gives the upper bound. $\square$

LEMMA D.2 (CORRELATED BERNOULLI SQUARE-ROOT COMPARISON). *Let I_i be arbitrarily dependent indicators and let H_i be independent indicators. If $\mathbb{E}I_i \leq \beta \mathbb{E}H_i$ for every i and $\beta \geq 1$, then for all $a_i \geq 0$,*

$$\mathbb{E} \sqrt{\sum_i a_i I_i} \leq \frac{\beta}{1 - e^{-1}} \mathbb{E} \sqrt{\sum_i a_i H_i}.$$

PROOF. Use $\sqrt{z} = (2\sqrt{\pi})^{-1} \int_0^\infty (1 - e^{-tz}) t^{-3/2} dt$. For fixed t , with $s_i = 1 - e^{-ta_i}$,

$$\mathbb{E}(1 - e^{-t \sum a_i I_i}) \leq \min \left\{ \sum_i s_i \mathbb{E}I_i, 1 \right\} \leq \beta \min \left\{ \sum_i s_i \mathbb{E}H_i, 1 \right\}.$$

Independence gives

$$\mathbb{E}(1 - e^{-t \sum a_i H_i}) = 1 - \prod_i (1 - s_i \mathbb{E}H_i) \geq 1 - e^{-\sum_i s_i \mathbb{E}H_i} \geq (1 - e^{-1}) \min \left\{ \sum_i s_i \mathbb{E}H_i, 1 \right\}.$$

Integrate. $\square$

LEMMA D.3 (COAREA FROM A SPECTRAL GAP). *Let P be a finite reversible Markov kernel with stationary law π and gap $\gamma > 0$. If (X, Y) has stationary one-step law, then*

$$\mathbb{E}_\pi |F - \mathbb{E}_\pi F| \leq \frac{4}{\gamma} \mathbb{E} |F(X) - F(Y)|. \quad (97)$$

If P is the two-block chain $X \rightarrow G \rightarrow Y$, then

$$\mathbb{E}_\pi |F - \mathbb{E}_\pi F| \leq \frac{8}{\gamma} \mathbb{E} |F(X) - \mathbb{E}(F(X) | G)|. \quad (98)$$

PROOF. For A with $\pi(A) \leq 1/2$, Poincaré applied to $\mathbf{1}_A$ gives $Q(A, A^c) \geq \gamma \pi(A) \pi(A^c) \geq (\gamma/2) \pi(A)$. Let m_F be a median. Integrate this inequality over superlevel sets of $(F - m_F)_+$ and $(m_F - F)_+$, using

$$|g(x) - g(y)| = \int_0^\infty |\mathbf{1}\{g(x) > t\} - \mathbf{1}\{g(y) > t\}| dt.$$

Reversibility turns the two directed boundary flows into $\mathbb{E}|g(X) - g(Y)|$, yielding $\mathbb{E}|F - m_F| \leq (2/\gamma)\mathbb{E}|F(X) - F(Y)|$. Since $\mathbb{E}|F - \mathbb{E}F| \leq 2\mathbb{E}|F - m_F|$, this is (97). In a two-block chain, X, Y are conditionally iid given G , so $\mathbb{E}|F(X) - F(Y)| \leq 2\mathbb{E}|F(X) - \mathbb{E}(F(X) | G)|$. $\square$

LEMMA D.4 (FINITE DOBRUSHIN COMPARISON). *Consider two conditional Gibbs laws on the same unpinning coordinates, whose boundary conditions differ only at coordinate i . If d_j is the TV distance of their j th marginals, then*

$$d \leq C_{-i,i} + C_{-i,-i}d, \quad d \leq (I - C_{-i,-i})^{-1}C_{-i,i}.$$

PROOF. Update the unpinned coordinates using maximal couplings. A disagreement at j can be caused directly by the boundary disagreement with probability at most C_{ji} or by a disagreement at k with contribution at most $C_{jk}d_k$. This gives the first coordinatewise inequality. Iterate it; the Neumann series converges because the relevant row and column norms are less than one. This is the finite binary specialization of the Dobrushin comparison theorem [10, Theorem 8.20]. $\square$

LEMMA D.5 (DOBRUSHIN LINEAR MGF). *Let X have a finite binary law with symmetric interdependence matrix C and $\|C\|_2 \leq 1 - \kappa$. Then for every $u \in \mathbb{R}^m$,*

$$\log \mathbb{E} \exp\{\lambda \langle u, X - \mathbb{E}X \rangle\} \leq \frac{\lambda^2 \|u\|_2^2}{2\kappa^2}. \quad (99)$$

The same bound holds after pinning and for every coordinate projection.

PROOF. The Dobrushin bounded-difference martingale inequality, in the normalization of Kontorovich and Raginsky [12, Theorem 4.2 and Section 5.1, Corollary 2], is

$$\log \mathbb{E} e^{\lambda(f - \mathbb{E}f)} \leq \frac{\lambda^2}{8} \|(I - C)^{-1} \delta(f)\|_2^2,$$

where $\delta_i(f)$ is the oscillation when spin i is changed. For $f(x) = \langle u, x \rangle$, $\delta_i(f) = 2|u_i|$. Symmetry gives $\|(I - C)^{-1}\|_{2 \rightarrow 2} \leq 1/\kappa$, proving (99). Pinning takes a principal submatrix. A projection is handled by extending u by zero before applying the full-law inequality. $\square$

LEMMA D.6 (PINNED GLAUBER MIXING). *Let K_ρ be random-scan heat-bath dynamics on an r -spin pinned Ising target satisfying (2). Then*

$$\max_x d_{\text{TV}}(K_\rho^T(x, \cdot), Q^\rho) \leq r \exp(-\kappa T/r).$$

PROOF. Couple neighboring configurations differing at j with the same updated coordinate and a maximal coupling of the two update laws. Updating j removes the disagreement. Updating $i \neq j$ leaves the old disagreement and creates a new one with probability at most C_{ij} . Hence expected Hamming distance after one step is at most

$$1 - 1/r + r^{-1} \sum_{i \neq j} C_{ij} \leq 1 - \kappa/r.$$

Concatenate these couplings along a shortest Hamming path, as in path coupling [5]. Couple an arbitrary start to stationarity; after T steps the expected distance is at most $r(1 - \kappa/r)^T$, which upper-bounds disagreement probability and TV. $\square$

The Hoeffding and Chernoff bounds used for amplification follow from the exponential Markov inequality and Hoeffding's lemma; the precise independent bounded-summand statement is [11, Theorem 2]. No optimal constants from these inequalities are used.

E AN ADAPTIVE-SCOND LOWER BOUND

We prove Theorem 3.6 directly. Let U be uniform on $\{-1, +1\}^n$. For $b \in \{-1, +1\}^n$ and $0 < \theta < 1/2$, define

$$P_b(x) = 2^{-n} \prod_{i=1}^n (1 + \theta b_i x_i). \quad (100)$$

This is the product Ising law with $J = 0$ and $h_i = b_i \operatorname{arctanh} \theta$. It has Dobrushin gap one and is hereditarily $1/4$ -balanced.

LEMMA E.1 (SEPARATION). *There are universal $a_0, c_0 > 0$ such that if $n\theta^2 \leq a_0$, then $d_{\text{TV}}(P_b, U) \geq c_0 \theta \sqrt{n}$ for every b .*

PROOF. Let $L_b = dP_b/dU$. Independence gives

$$\mathbb{E}_U L_b^2 = (1 + \theta^2)^n, \quad \mathbb{E}_U L_b^3 = (1 + 3\theta^2)^n, \quad \mathbb{E}_U L_b^4 = (1 + 6\theta^2 + \theta^4)^n. \quad (101)$$

Hence $\mathbb{E}(L_b - 1)^2 \geq n\theta^2$. The fourth centered moment is the linear combination

$$(1 + 6y + y^2)^n - 4(1 + 3y)^n + 6(1 + y)^n - 3, \quad y = \theta^2.$$

It and its first derivative vanish at $y = 0$; on $0 \leq y \leq a_0/n$, its second derivative is at most Cn^2 by direct differentiation. Thus $\mathbb{E}(L_b - 1)^4 \leq Cn^2\theta^4$. Log-convexity of L^p norms, equivalently Hölder, gives

$$\mathbb{E}|L_b - 1| \geq \frac{(\mathbb{E}(L_b - 1)^2)^{3/2}}{(\mathbb{E}(L_b - 1)^4)^{1/2}} \geq c_0 \theta \sqrt{n}.$$

Divide by two and adjust c_0 . □

Choose a universal A with $c_0 A \geq 2$ and set

$$\theta = A\varepsilon/\sqrt{n}. \quad (102)$$

For $\varepsilon \leq \varepsilon_0$ small enough, Lemma E.1 makes every P_b ε -far from U and ensures $\theta < 1/2$.

Now put the uniform prior on b and consider s ordinary samples. The chi-square divergence of the resulting mixture from $U^{\otimes s}$ is

$$\begin{aligned} 1 + \chi^2 &= \mathbb{E}_{b, b'} (\mathbb{E}_U L_b L_{b'})^s \\ &= \left(\frac{(1 + \theta^2)^s + (1 - \theta^2)^s}{2} \right)^n. \end{aligned} \quad (103)$$

For $y = \theta^2 \leq 1/2$,

$$\frac{(1 + y)^s + (1 - y)^s}{2} = (1 - y^2)^{s/2} \cosh(s \operatorname{arctanh} y) \leq e^{2s^2 y^2},$$

using $\cosh z \leq e^{z^2/2}$ and $\operatorname{arctanh} y \leq 2y$. Therefore

$$1 + \chi^2 \leq \exp(2ns^2\theta^4). \quad (104)$$

If $s \leq c\sqrt{n}/\varepsilon^2$ for a sufficiently small universal c , (102)–(104) make the TV distance between the mixture transcript and the null transcript less than $1/6$.

It remains to upgrade the ordinary-sample obstruction to fully adaptive SCOND. For a fixed product law P_b , one SCOND response to a restriction ρ can be generated from one fresh unconditional $X \sim P_b$: retain X on every free coordinate and overwrite each pinned coordinate by its requested value. Product independence makes the result exactly $P_b(\cdot \mid \rho)$. The same simulator is exact under U , uses no knowledge of b , and can be run online after every adaptively chosen query. Thus an s -query adaptive SCOND tester would induce an s -ordinary-sample test between the null

and the preceding mixture. Data processing and Yao's principle contradict the $1/6$ transcript bound if both errors are at most $1/3$. This proves $s = \Omega(\sqrt{n}/\epsilon^2)$.

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